Testing mean reversion in Reliance

Daily price signals and a cost-aware long-only backtest

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Executive assessment

This study tests whether unusually low prices relative to a recent average offer profitable rebound opportunities in RELIANCE.NS. The fixed baseline earns **2.70%** in development and **−0.56%** in the 2024 final test after simulated costs. Buy-and-hold returns are 428.52% and −5.64%, respectively. The strategy loses less in 2024, but its very low average exposure and only three completed test trades prevent a strong claim of trading skill.

Hypothesis and indicator

The hypothesis is that some short-term price dislocations reverse. It can fail when adverse information changes the price level or a downward trend persists. A trailing 20-session mean provides a local reference; a Z-score standardises the deviation by recent price dispersion. These are fixed starting choices, not estimated optimal parameters.

$\boldsymbol{\mu_t}$ = mean of $C_{t-19}, \ldots, C_t$; $\boldsymbol{s_t}$ = sample standard deviation of those 20 closes;
$\boldsymbol{Z_t} = (C_t - \mu_t) / s_t$, with ddof = 1 [3]. A value below −2 means a price deviation exceeding two rolling standard deviations; it does not establish a rebound probability or that price levels are normally distributed.

Component	Fixed baseline specification
Entry	If flat and the preceding session's Z < −2, buy at the current session's adjusted Open with adverse slippage.
Exit	If holding and the preceding session's Z ≥ 0, sell at the current adjusted Open with adverse slippage.
Sizing	Initial cash INR 100,000; spend at most 10% of available cash including entry fee; integer adjusted-price units. One long position; no leverage.
Trading costs	Slippage 0.05% against each fill; fee 0.10% on each buy and sell. Idle cash earns zero.
Baseline risk control	Entry size only. No stop-loss or time limit; a losing position may remain open indefinitely.

Execution and cash accounting

Let s = 0.0005 and f = 0.001. The buy fill is Open × (1 + s), and quantity is floor[0.10 × cash / (buy fill × (1 + f))]. The sell fill is Open × (1 − s), with net proceeds equal to quantity × sell fill × (1 − f). This charges fees on both sides and includes entry fees inside the cash budget. The quantity floor enforces integer research units.

Data and scope

Yahoo Finance via yfinance supplies 2,221 daily sessions spanning 2016-01-01 to 2024-12-31 [1,2]. Reliance offers a long history for a single-stock case study. Daily OHLCV supports close-based signals and simulated next-open fills, but cannot establish order-book liquidity or actual fill quality.

Checks cover missing/non-finite values, duplicate rows, non-positive prices and invalid OHLC ordering. Suspicious data trigger an error rather than silent deletion; extreme returns are retained absent evidence of error. Adjusted OHLC is internally consistent for this research simulation, with material execution and sizing limitations discussed on page 6.

Performance metrics and accounting conventions

Development covers the first session of 2016 through the last session of 2022. The final test uses 29 December 2023 as its equity baseline and ends 31 December 2024. Each standalone run starts flat with INR 100,000. Earlier prices provide trailing indicator history; orders use the previous session's signal only.

Metric	Development 2016-2022	Final test 2024
Cumulative return (%)	2.70	−0.56
Annualised return or CAGR (%)	0.38	−0.56
Sharpe ratio	0.28	−0.61
Sortino ratio	0.40	−0.85
Maximum drawdown (%)	−3.32	−1.31
Win rate (%)	75.00	33.33
Completed trades	24	3
Profit factor	1.97	0.41
Average winning trade (INR)	293.10	337.36
Average losing trade (INR)	−446.06	−409.44
Largest winning trade (INR)	714.11	337.36
Largest losing trade (INR)	−1,685.46	−794.44
Buy-and-hold return (%)	428.52	−5.64
Average invested exposure (%)	1.68	2.42
Ending marked equity (INR)	102,703.70	99,436.11
Open adjusted-price units	8	8
Estimated liquidation return (%)	2.69	−0.58

Portfolio returns and risk metrics

Marked equity is cash plus held units valued at the adjusted close. Daily return is $r_t = E_t / E_{t-1} - 1$. Cumulative return is $E_{end} / E_{start} - 1$; CAGR uses actual elapsed days divided by 365.25. Drawdown is $E_t / \max(E_0, \ldots, E_t) - 1$; maximum drawdown is its most negative value. Trading costs and idle cash are included.

Sharpe = $\sqrt{252} \times$ mean(r) / sample standard deviation(r), assuming a zero annual risk-free rate. Sortino uses the same numerator and $\sqrt{\text{mean}[\min(r, 0)^2]}$, computed over **all** daily observations, with zero minimum acceptable return. These are conventional annualisations; serial dependence and a short test window limit their interpretation.

Completed trades and unfinished positions

Trade P&L is net of entry and exit costs. Win rate counts profitable completed trades; profit factor divides total realised gains by the absolute total realised losses. Open-position P&L enters equity metrics but not completed-trade statistics. Estimated liquidation is a separate last-close scenario, not a fabricated strategy exit.

Undefined ratios are N/A when a denominator is zero or the relevant wins, losses or trades do not exist. Profit factor is N/A with no losing trades. All displayed metrics reproduce the original study's rounded values [4].

Development behaviour and validation integrity

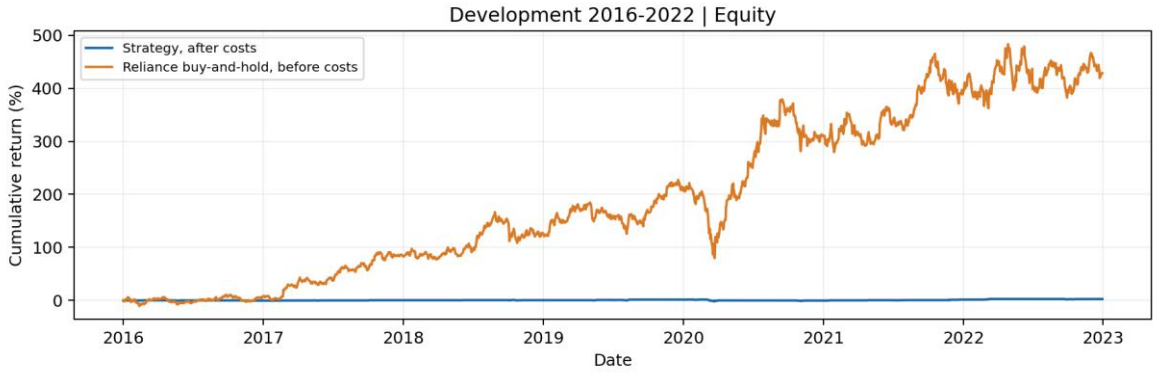


Figure 1. Development cumulative return, 2016-2022. Buy-and-hold participates in the large price rise; the mostly cash strategy captures little of it. Lines are percentage returns from their initial equity, not rupee balances.

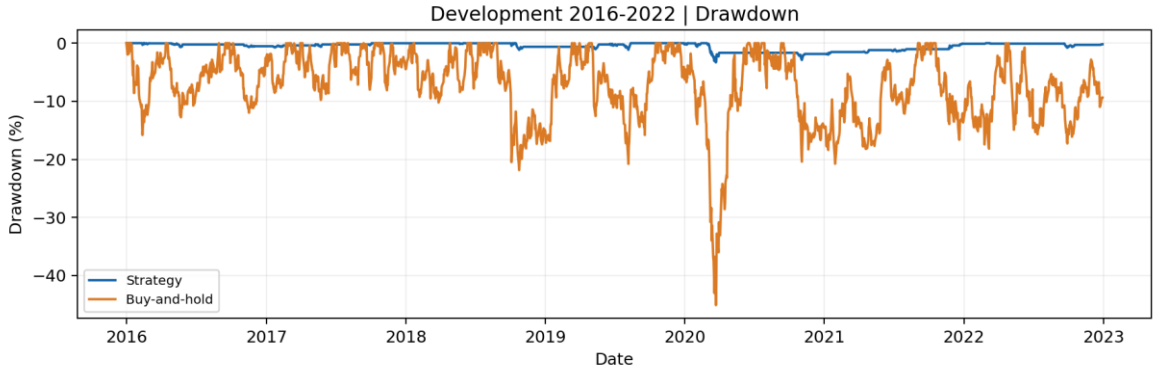


Figure 2. Development drawdown. The strategy's small drawdown accompanies average invested exposure of only 1.68%. It cannot be attributed to an independently effective exit rule.

What the development result shows

The 2.70% cumulative gain corresponds to only 0.38% annualised, despite a 75% win rate across 24 completed trades. The average loss of INR 446.06 exceeds the average win of INR 293.10, and the largest loss is INR 1,685.46. Profitability therefore depends on the frequency of winners and the size of occasional losses, rather than win rate alone.

Using the rounded win/loss averages, completed-trade expectancy is approximately **INR 108.31 per trade**: 0.75 × 293.10 − 0.25 × 446.06. This excludes the unfinished position and does not equal total portfolio return. Low exposure and a long idle-cash period help explain why positive trade expectancy translates into a small portfolio gain.

How the 2023 validation result should be read

The reported 2023 validation slice is part of a **continuous 2016-2023 run**, not a fresh INR 100,000 portfolio. It records 0.26% strategy return, 9.11% benchmark return and −0.75% drawdown, with five completed trades entered in 2023. Starting holdings and cash may carry forward, so it should not be treated as an independent reset backtest.

Original exploration included validation data; 2023 was therefore not fully unseen. The separately labelled 2024 baseline was evaluated after its rules were fixed. Later parameter and risk-control experiments use development data but were proposed after viewing the baseline test outcome. They remain exploratory and cannot inherit an untouched 2024 test.

Final test execution and benchmark comparison

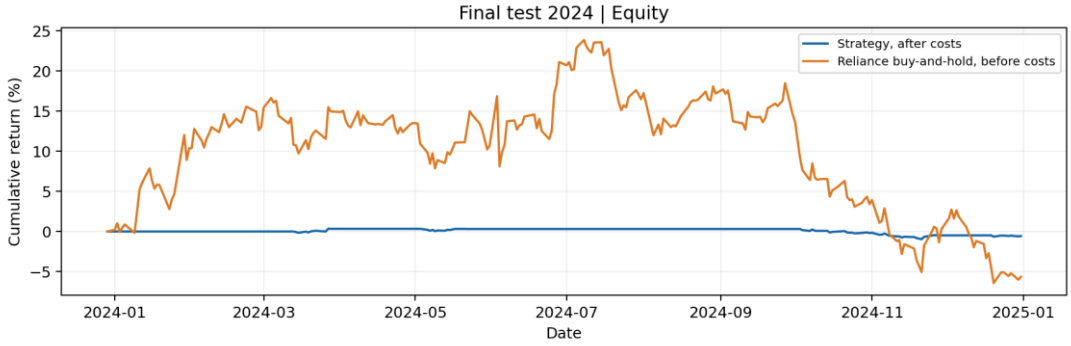


Figure 3. Final-test cumulative return. The strategy ends at −0.56%, versus −5.64% for the fully invested same-stock benchmark. The difference is approximately +5.08 percentage points.

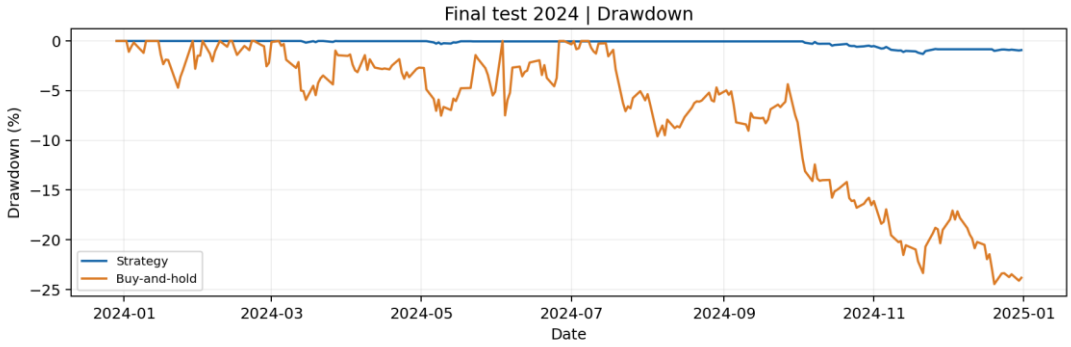


Figure 4. Final-test drawdown. Strategy maximum drawdown is −1.31%, but average invested exposure is only 2.42%. The comparison reflects different market exposure as well as trading decisions.

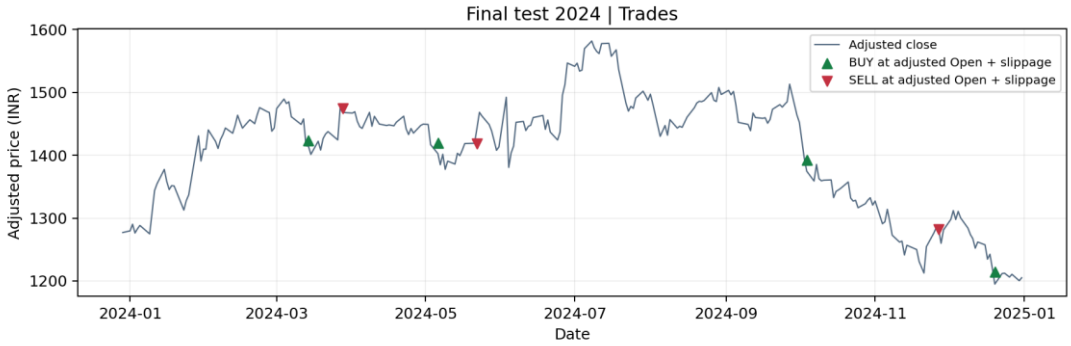


Figure 5. Adjusted price with simulated entry and exit fills. Markers include an entry that remains open at year-end. Signals from the prior close are filled at the next adjusted open plus or minus slippage.

The result supports caution rather than an edge claim

Only three trades finish in the test: one win and two losses, giving a 33.33% win rate and profit factor of 0.41. One additional outcome would materially alter these statistics. Ending marked equity is INR 99,436.11 with eight adjusted-price units still held; estimated liquidation changes return to −0.58%. The strategy loses money even though it declines less than buy-and-hold.

Holding Reliance is a relevant reference for the opportunity cost of timing this stock. It is not a broad-market comparison or a clean isolation of timing skill: the benchmark is fully invested and before costs, while the strategy is mostly cash and after costs. An exposure-matched, cost-matched benchmark and cash comparison are proposed next checks, not results already established.

Parameter sensitivity market regimes and exits

The following checks use development data and the same trading-cost assumptions. They were added after the baseline final-test result was known. The original 20-session, −2-entry rule remains the baseline; selecting the best-looking row now would be retrospective optimisation.

Window	Entry Z	Return %	Max DD %	Trades
15	−2.5	1.87	−2.35	7
15	−2.0	5.77	−3.46	37
15	−1.5	8.40	−3.94	53
20	−2.5	1.09	−3.34	13
20	−2.0	2.70	−3.32	24
20	−1.5	3.87	−3.51	41
25	−2.5	0.48	−3.32	10
25	−2.0	3.98	−3.49	23
25	−1.5	3.00	−3.49	32

Returns range from 0.48% to 8.40%, showing meaningful parameter sensitivity. Within each window, a less negative entry threshold increases trade count. Return does not improve monotonically: with a 25-session window, −1.5 yields 3.00%, below 3.98% at −2.0. This grid diagnoses sensitivity; it does not demonstrate stable performance outside development.

Behaviour conditional on the preceding trend

Each day is labelled using the previous session's trailing 60-session stock return: rising above +5%, falling below −5%, and sideways otherwise. Warm-up observations are excluded. The labels describe the preceding trend, not the direction of the current day's return.

Prior regime	Sessions	Mean daily strategy %	Mean daily benchmark %
Falling	271	0.0088	0.3164
Rising	806	0.0011	0.1297
Sideways	592	−0.0015	0.0018

Positive benchmark returns on "falling" days are therefore possible. These arithmetic averages pool non-contiguous observations; they are not compounded regime returns or separate investable portfolios. The strategy's small values also reflect low exposure, so they do not establish a profitable regime filter.

Explicit exit-control experiment

A separate variant adds a 5% loss trigger using the preceding adjusted close relative to entry fill and a maximum 20 held sessions. Exits occur at the next open with costs. An overnight gap can exceed the 5% threshold; this is not a guaranteed intraday stop.

Development variant	Return %	Max DD %	Trades
Baseline sizing only	2.70	−3.32	24
5% close-stop plus 20 sessions	1.56	−3.95	28

The variant reduces return by 1.14 percentage points and worsens maximum drawdown by 0.63 percentage points. Not adopted as the baseline; other stop-loss rules may still work.

Limitations improvements and reproducibility

What the backtest establishes

The study implements an interpretable mean-reversion rule with delayed execution and explicit costs. It finds modest development gains, weak capital utilisation and a negative final-test return. The evidence is insufficient to establish a durable advantage. The main contribution is a transparent test of the idea and an explanation of why apparently reassuring metrics can be misleading.

Material modelling limitations

Adjusted-price execution. Retrospectively adjusted OHLC is not a reconstruction of historical executable rupee prices, share quantities, splits and dividend cash flows. Integer sizing on adjusted prices can depend on later adjustment factors. The traded quantities are research units; live implementation requires raw fills and explicit corporate-action accounting.

Risk and exit interpretation. Spending 10% of cash limits initial allocation, not a fixed loss percentage. Exposure can change with prices, and the baseline has no stop or holding limit. Returning to a moving average need not produce a gain: the average itself may fall, while next-open gaps and costs can erase a rebound.

Generalisability and costs. One stock, a small final-test trade sample and fixed fee/slippage assumptions limit confidence. No market impact, spread variation, interest on cash or uncertainty interval is modelled. The approximate 0.30% two-sided fee-plus-slippage burden, before compounding, is an assumption rather than a verified brokerage schedule. Low exposure also limits comparison with fully invested buy-and-hold.

Next research steps

1. Rebuild execution with raw prices, explicit corporate actions and a frozen point-in-time snapshot.
2. Compare cash, exposure-matched and cost-matched benchmarks, then test sensitivity to worse fills and higher costs.

3. Evaluate pre-specified rules across more stocks and chronological development folds.

4. Reserve genuinely unseen data for the final decision; do not retune to the observed 2024 loss.
These are proposed experiments, not additional completed backtests.

Reproduction and assistance

The original submission records a dated adjusted-price CSV, data fingerprint, actual package versions, generated metrics CSVs, requirements.txt and retained analysis cells [4]. A local snapshot is preferable for exact reproduction because a fresh provider download may change the data. This report revision preserves the reported backtest and charts; it does not independently rerun the notebook. ChatGPT/Codex assistance with explanations, code, debugging, review, metrics, charts and report preparation is acknowledged. The author remains responsible for reviewing and explaining the submission.

Recorded data SHA-256
082924f5bea83a2acc0ac3460040b23c87c782acc40ab52a4831f06cf77ace42

Resources and evidence

[1] Yahoo Finance. RELIANCE.NS historical prices. Source link.

[2] yfinance. Download API, including OHLC adjustment and date-boundary conventions. Source link.

[3] pandas. Rolling standard deviation, ddof=1. Source link.

[4] CFA Club, IIT Indore. Induction Task 2026, Finance and Quant, pp. 5-8; supplied task PDF.

[5] Software documentation cited in the original study: pandas, NumPy, Matplotlib and ReportLab .